

KnowSelf: Teaching LLM Agents When to Think, Reflect, or Seek Knowledge



Agentic knowledgeable self-awareness for planning systems

Key idea

LLM agents autonomously regulate knowledge use via situational tokens.

Three modes: act directly • reflect • retrieve external knowledge.

Source: arxiv.org/abs/2504.03553



Current Agent Training Uses “Flood Irrigation”

Indiscriminate injection of trajectories/feedback/knowledge hurts efficiency and robustness

Trajectory imitation

Example: ReAct

- Learns from gold trajectories
- Can be brittle pattern-fitting
- Fragile under unexpected inputs

Trial-and-error refinement

Example: Reflexion

- Uses external feedback
- High inference cost
- Not situationally applied

Knowledge-augmented planning

Examples: KnowAgent, WKM

- Injects domain knowledge broadly
- Often 100% knowledge usage
- Unnecessarily high inference cost

Flood irrigation

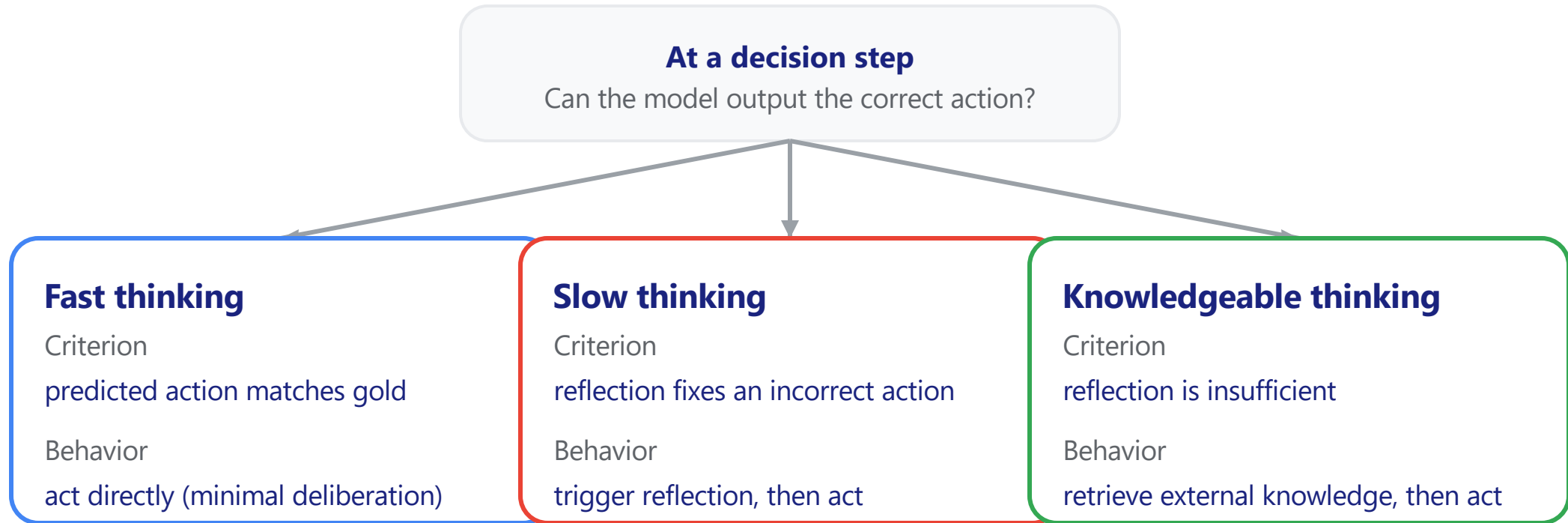
Always inject extra supervision or knowledge
→ wasteful; can cause fragile planning
→ high inference cost

Selective irrigation (KnowSelf)

Learn situational self-awareness tokens
→ reflect or retrieve only when needed
→ lower cost, higher robustness

Three Situational Modes: Fast, Slow, Knowledgeable Thinking

A decision-time policy: act directly, reflect, or retrieve knowledge

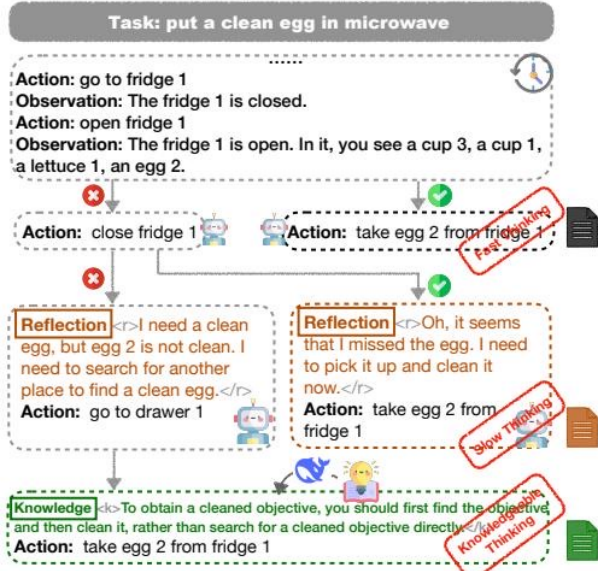


KnowSelf Framework: Data → Training → Self-Aware Inference

Special tokens trigger direct action, reflection, or knowledge retrieval

Step 1

Self-awareness Data Construction



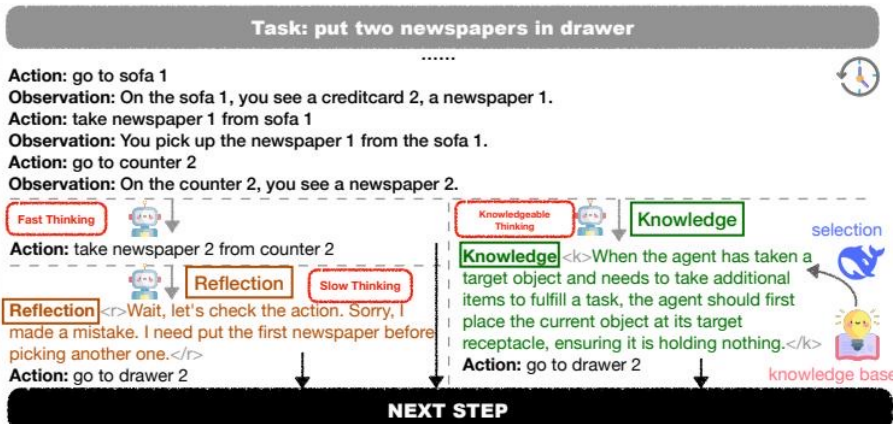
Step 2

Self-awareness Learning



Step 3

Self-awareness Inference



1) Data construction

Self-explore; heuristic situation judgment; insert special tokens per step.

2) Two-stage training

SFT learns token patterns;
RPO refines token decisions.

3) Self-aware inference

Generated tokens trigger
act / reflect / retrieve knowledge.

Qualitative Examples: Reflection vs. Knowledge Tokens

KnowSelf resolves failures by triggering the right cognitive mode

(a) Task: put a hot mug in cabinet	
OpenAI-O1 Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. [Knowledge] <k>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</k> Action: take egg 1 from microwave 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. Action: put mug 1 in/on microwave 1 [Reflection] <r>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</r> Action: heat mug 1 with microwave 1
(b) Task: put a saltshaker in drawer	
DeepSeek-R1 Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is open. In it, you see a fork 1. Action: put saltshaker 1 in/on drawer 1 [Reflection] <r>Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</r> Action: close drawer 1 Observation: You close the drawer 1. Action: put saltshaker 1 in/on drawer 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is closed. Action: open drawer 1 Observation: You open the drawer 1. In it, you see a fork 1. [Knowledge] <k>When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</k> Action: put saltshaker 1 in/on drawer 1

A) “put a hot mug in cabinet”

KnowSelf uses a Reflection token to heat the mug via the microwave.

B) “put a saltshaker in drawer”

KnowSelf uses a Knowledge token to place the object without detours.

ALFWorld Results: Success Rate (%) with Know% Usage

KnowSelf achieves 84.33% (Llama-8B) using only 15.01% external knowledge

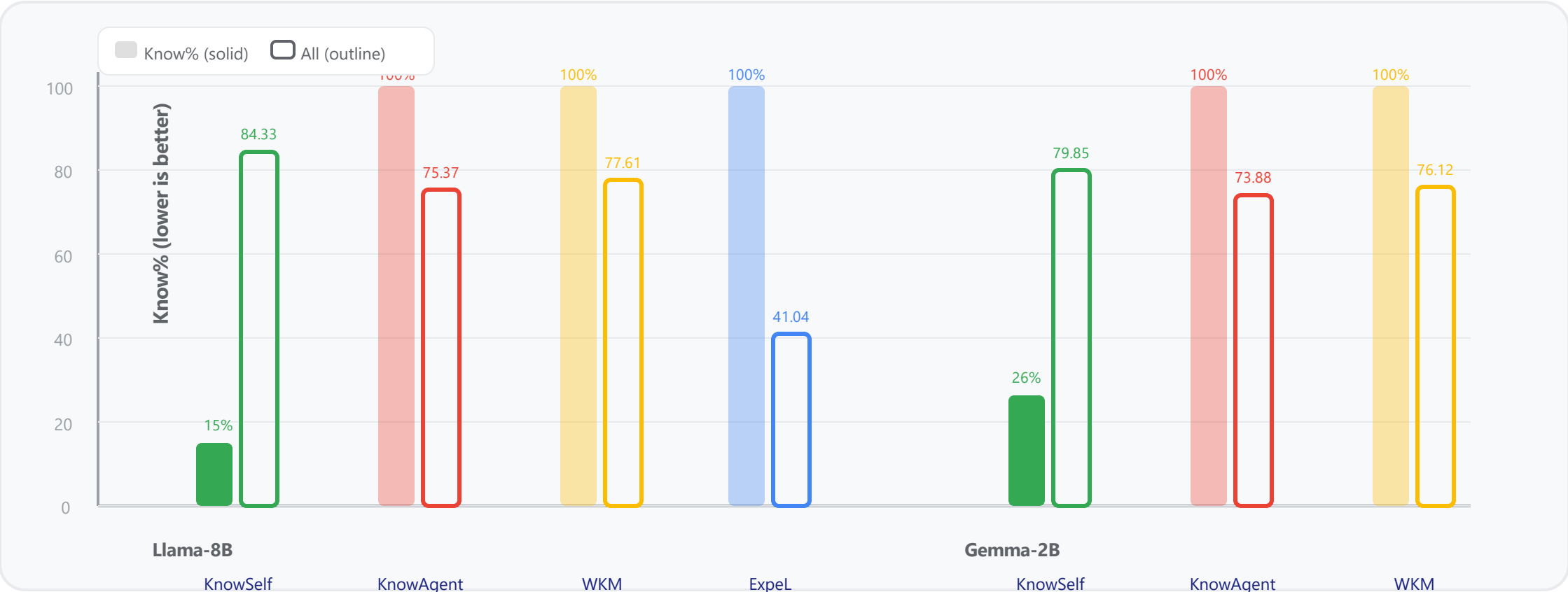
Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Know% indicates proportion of steps using external knowledge.



Minimal Knowledge, Maximum Impact: Efficiency + Scaling

KnowSelf uses ~15–26% knowledge vs 100% baselines, while achieving top accuracy



Takeaway: selective retrieval can dominate full-knowledge prompting.



Key Takeaways: Self-Awareness Is the Missing Piece

Learn when to act, reflect, or retrieve — and save knowledge calls

Findings

- 1) Flood irrigation knowledge is wasteful/suboptimal.
- 2) Situational self-awareness via special tokens + lightweight heuristic labeling.
- 3) Llama-8B KnowSelf: 84.33 vs GPT-4o ReAct: 76.12 and GPT-4o ExpeL: 79.85.
- 4) Only 15–26% knowledge usage vs 100% competitors.
- 5) Scales down: Gemma-2B KnowSelf reaches 79.85, matching GPT-4o ExpeL.

Future directions

- Calibrate “need knowledge” triggers more precisely.
- Extend beyond ALFWorld-style planning tasks.
- Broader knowledge sources + safety-aware retrieval.
- Better robustness under distribution shifts.

Closing message

Situational self-awareness is a control layer for agent cognition: decide when to think and when to retrieve knowledge.

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